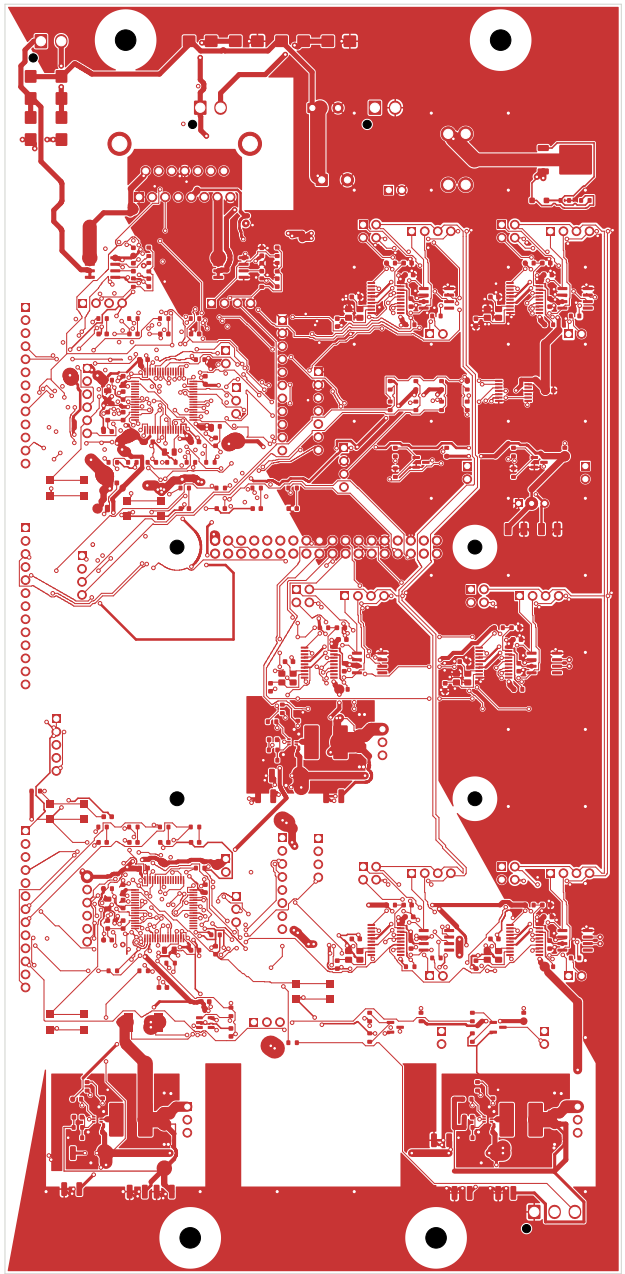
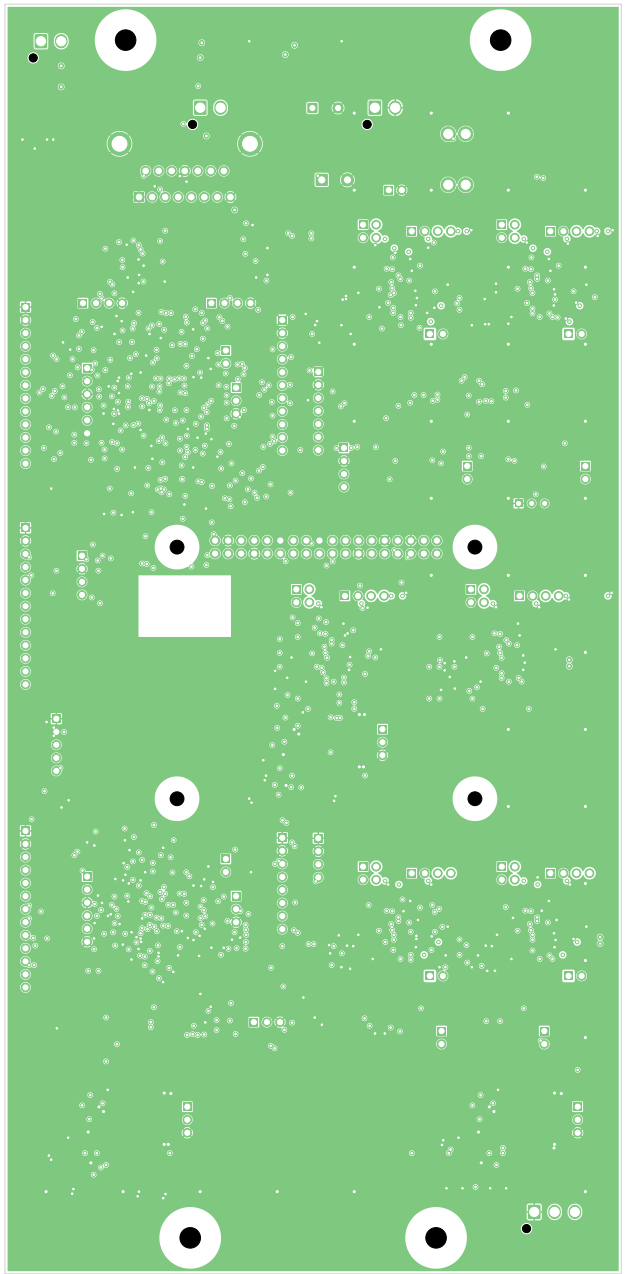
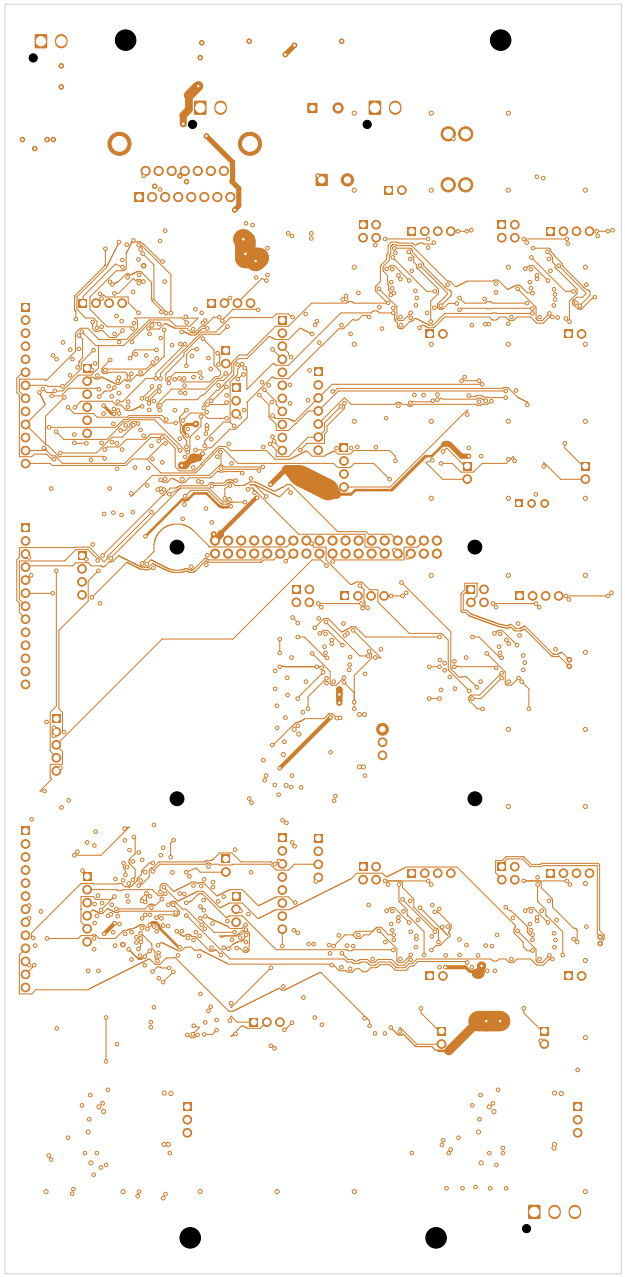
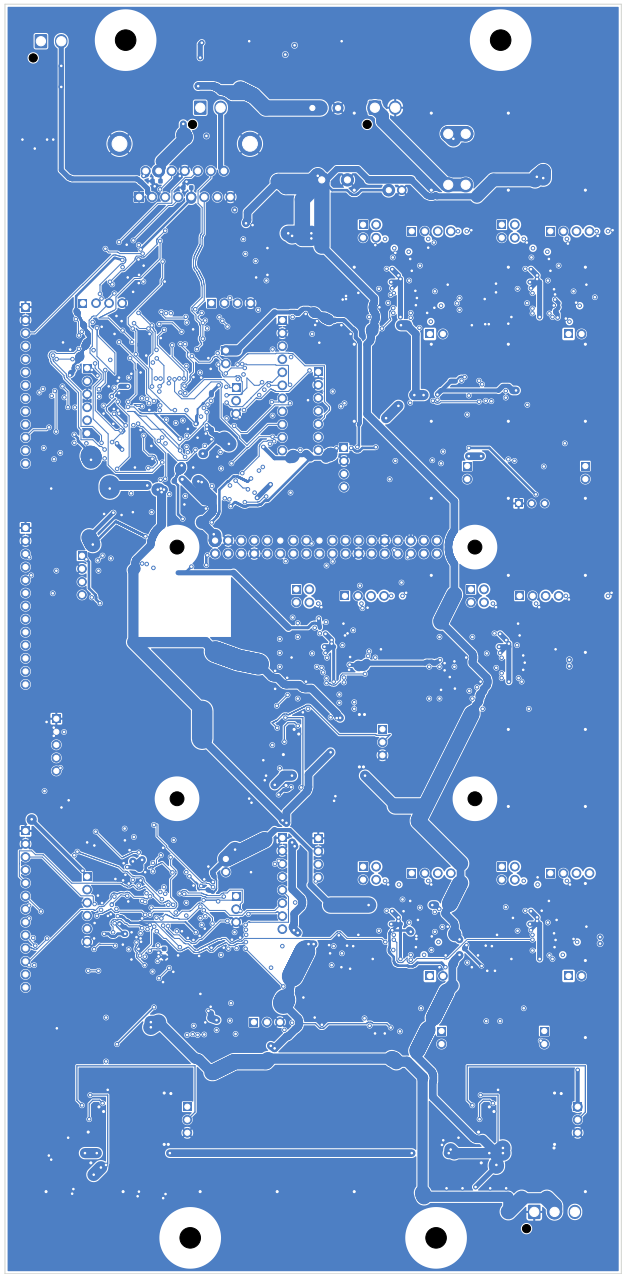
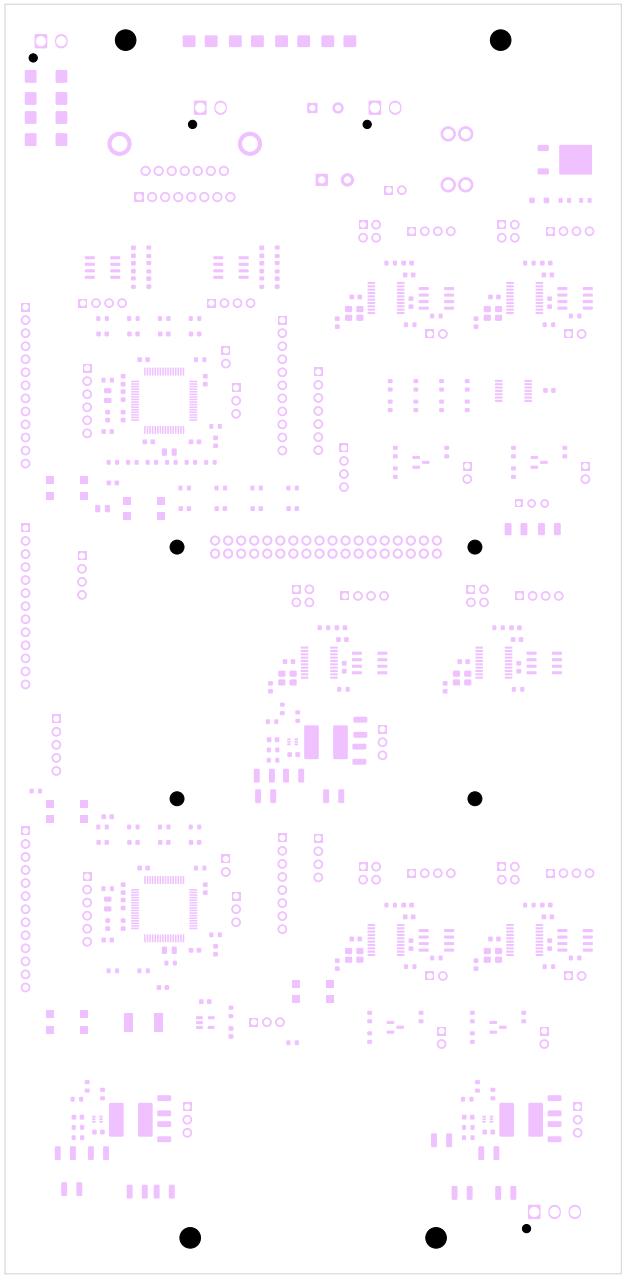
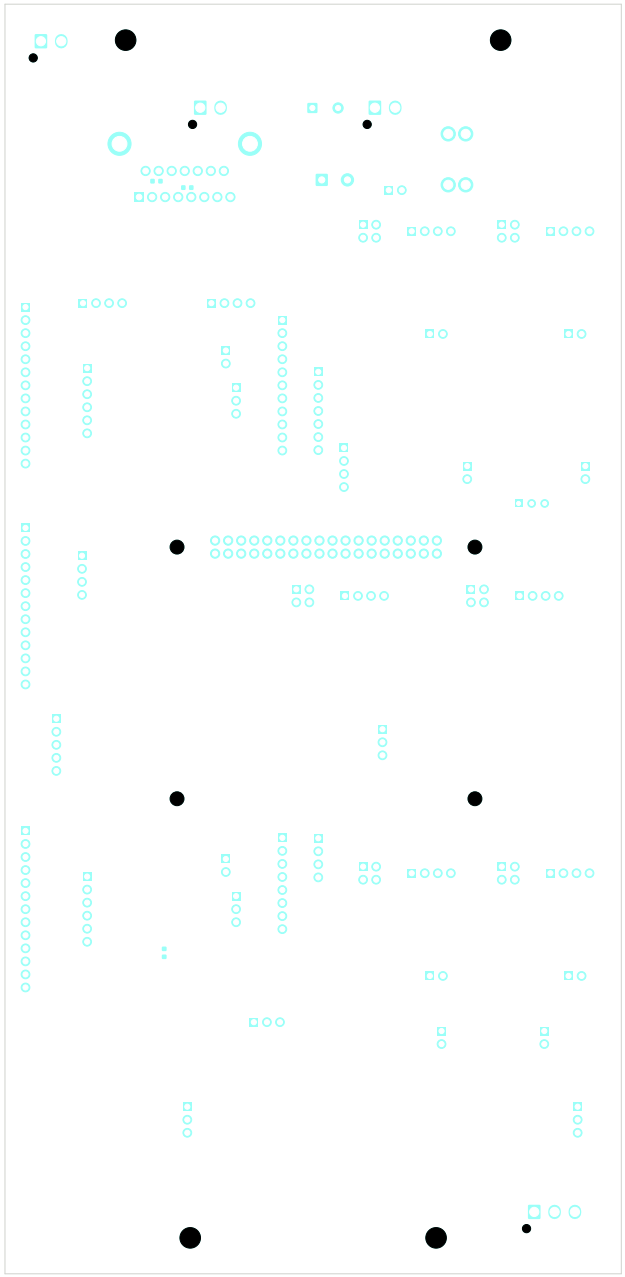












649 Team 4
H1
H2
H3
H4
J80 L
J81 MOTOR R
J40 1:BAT+ 2:GND J40 BAT IN
J41 1:VBAT 2:GND
J41
VBAT / GND
RX
J21 REAR CAN A TEST
J25 REAR CAN B TEST
TX RX H L
J20 REAR CAN A ISO
J24 REAR CAN B ISO
J603
REAR TIMING
GND
RXB
TXA
TXB
DEGA
DEGB
SET
DIR
RAW ADC
J2
MOTOR L
J3 MOTOR R
SCL GND
J601 REAR SWD
3V3
CLK
GND
DIO
RST
GND
J600 REAR BOOT
3V3
BOOT
TX
RX
GND
STEER
PID
RPWM
J602
REAR UART
RADC
J604 REAR PROBES
GND
GND
LA
LB
RA
RB
J605
REAR EXTRA
3V3
A
B
J83
ENC R
JP20 TERM A REAR
JP24 TERM B REAR
LED+
SINK
J650 REAR L LED
J651 R LED
SW60
REAR RESET
SW61
REAR SELF
2
1
J70 PI GPIO (BOTTOM)
C804
C805
GND
GND
UDP
3V3
J82
ENC L
GND
A
TXB
B
J703
PI TIMING
NC
NC
NC
NC
NC
J30 PI CAN A ISO
J31 PI CAN A TEST
J34 PI CAN B ISO
J35 PI CAN B TEST
TX RX H L
J43 5V1 / PG / G
PG
J10//20//30//34: 1/2=H, 3/4=L (ISO)
J21: 1:TX 2:RX 3:H 4:L
GND
GND
SELF
ERR
STAT
J704
PI PROBES
SW70 PI SELF
J505 FRONT EXTRA
J14 FRONT CAN B ISO
J15
FRONT CAN B TEST
GND
GND
RXA
RXB
TXA
TXB
DEGA
DEGB
FL
FR
RELAY
ERROR
J501 FRONT SWD
3V3
CLK
GND
DIO
RST
GND
J500
FRONT BOOT
3V3
BOOT
TX
RX
GND
TASK
STEER
PID
SADC
J502 FRONT UART
J10 FRONT CAN A ISO
J11 FRONT CAN A TEST
SET
DIR
J504
FRONT PROBES
H H
TX RX H L
R580
R581
C580
SW51 FRONT SELF
JP10 TERM A FRONT
JP14 TERM B FRONT
SW50 FRONT RESET
J503 FRONT TIMING
J1 SERVO
GND
RAW
ADC
LED+
SINK
J550 FRONT L LED
J551 FRONT R LED
PIN MAP: count from square pin 1; G = GND
J42 SERVO / PG / G
6V6
PG
GND
J603 2:G 3:RXA 10:LPWM 12:RL 13:RR
J703 4:CMD 5:TXA 7:DEGA 8:DEGB
J704 1:G 2:G 3:SELF 4:ERR
J604 1:G 2:G 3:SELF 4:ERR 5:TASK 9:OVLO 10:LADC
J505 1:G 2:G
J605 7:RELAY
J83 2:G
J2 2:SCALED
J503 9:SERVO
6V6
GND
J44 3V3 / PG / G
3V3
PG
GND
J90 SERVO
1:GND 2:6V6 3:PWM

1:M1 2:M2

Tianyi He
Felipe Marples Rodriguez
Sida Cheng

1:M1 2:M2

J0: ALIGN FIRST PAIR TO PI 3/4
PI 1/2 AWG 30/40 LEFT EXPOSED

